Uganda Institute of Information and Communications Technology

"In Quest of Excellence"

BSM121: RESEARCH METHODS

MODULE 3: Research Process

Instructor: Abok Tonny

Introduction

- The research process consists of a series of systematic steps that a researcher must go through in order to generate knowledge.
- By following an organized sequence of steps, researchers can:
  - Generate reliable data
  - Form valid conclusions
  - Contribute valuable knowledge to their field
- From defining a research question to reporting findings, each step of the process plays a critical role in producing high-quality, impactful research.
- Irrespective of the category of a research study, the steps followed in conducting it are the same.

An Overview of the Research Process

Step 1: Selecting and Defining Problem

- The first stage in conducting research is **identifying and clearly defining the research problem.**

- Definitions of research problem:

  - A research problem refers to an issue, gap in knowledge, or situation that requires investigation.

  - A problem is a gap between reality (present situation) and expectation (more desirable situation).

- Activities Involved

  - Observing real-world problems
  - Reviewing existing knowledge
  - Identifying gaps in literature
  - Defining the scope of the study

Step 1: Selecting and Defining Problem

- **Example:** A lecturer may observe that many university students fail to complete their studies, which could lead to research on factors influencing student dropout in higher education institutions.

- A well-defined research problem will guide the researcher through all stages of the research process, from setting objectives to choosing a technique.

Step 2: Conduct a Literature Review

- Once a problem has been found, the researcher conducts a literature search by examining existing research studies, books, journals, reports, and other scholarly materials related to the research problem.

- **Purpose**

  - To understand what has already been studied
  - To avoid duplication of research
  - To identify research gaps
  - To build theoretical and conceptual frameworks

- **Tips**:
  - Use reliable sources, such as peer-reviewed journals, books, government reports, institutional publications, and academic databases.
  - Organize findings that summarizes existing knowledge and identifies gaps for further study.

Step 3: Formulate Research Objectives and Questions

- After reviewing literature, the researcher defines the objectives and research questions that guide the study.

- **Research Objectives**
  - These describe what the study intends to achieve.
  - Example: To examine factors influencing student dropout.

- **Research Questions**
  - These guide the investigation.
  - Example: What factors contribute to student dropout in higher education institutions?

- **Hypotheses (if applicable)**
  - In quantitative research, researchers may formulate testable hypotheses.
  - Example: Financial challenges significantly influence student dropout.

Step 4: Choose the Research Design

- The research design is the **overall plan or blueprint** for conducting the research.
- It specifies how data will be collected, measured, and analyzed.
- **Components of research design include:**
  - Type of research (qualitative, quantitative, mixed)
  - Sampling techniques
  - Data collection methods
  - Data analysis procedures
- **Example:** A researcher studying student dropout may use a survey research design to collect data from students.

Step 5: Data Collection

- This stage involves gathering information relevant to the research objectives.
- Data may be obtained from **primary or secondary sources**.

- Methods of Data Collection
  - Surveys and questionnaires
  - Interviews
  - Observations
  - Experiments
  - Document analysis

- Example: A researcher may distribute questionnaires to university students to collect data about academic, financial, and social factors affecting their studies.

Step 6: Data Analysis (Results)

- Once data has been collected, it must be organized, processed, and analyzed in order to identify patterns, trends, or relationships in order to generate meaningful insights

- Steps Involved
  - Data entry
  - Data cleaning e.g. checking for missing data, duplicates, incorrect formats
  - Data coding: converting categories to numbers e.g. male 1, and female 0
  - Statistical analysis

- Tools Used: Researchers often use statistical software such as:
  - R programming language
  - Python
  - Stata
  - SPSS

Step 7: Interpretation of Results (Discussion)

- After analyzing the data, the researcher interprets the findings to determine what the results mean in relation to the research objectives and questions.

- Activities

  - Explaining patterns in the data
  - Comparing results with previous studies
  - Discussing implications of findings

- Example: If results show that financial constraints significantly affect student retention, the researcher explains the significance of this finding.

Step 8: Draw Conclusions and Make Recommendations

- The researcher summarizes the main findings and draws conclusions based on the evidence obtained from the analysis.

- Activities
  - Summarizing key results
  - Identifying implications for policy or practice
  - Providing recommendations for improvement

- Example: A study on student dropout:
  - Indicate that student drop out is attributed to financial difficulty (73.7%), academic performance (57.9%), and motivation (31.6%).
  - Recommendations may include
    - Increased financial support programs
    - Academic counseling services

Step 9: Report and Communicate Results

- This involves documenting and communicating research findings through a research paper, report, or presentation.
- Purpose
  - To share findings with the academic community
  - To inform decision-makers and policymakers
  - To contribute to knowledge
- Tips:
  - Follow formatting guidelines for research report for institutions, conferences or journals.
  - Use visuals like tables and charts to present data clearly.